

CBSE Class 12 Physics — Semiconductor Electronics

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQ/Assertion-Reason/VSA)

Q1. The formation of the depletion region in a $p - n$ junction diode is fundamentally due to: (a) movement of dopant atoms (b) diffusion of both electrons and holes (c) drift of electrons only (d) drift of holes only [PYQ 2023] Very High

Q2. Which one of the following elements will require the highest energy to take out an electron from its lattice? (a) **Ge** (b) **C** (c) **Si** (d) **Pb** [PYQ 2023] Very High

Q3. In an extrinsic semiconductor, the number density of holes is $4 \times 10^{20} \text{ m}^{-3}$. If the number density of intrinsic carriers is $1.2 \times 10^{15} \text{ m}^{-3}$, the number density of electrons in it is: (a) $1.8 \times 10^9 \text{ m}^{-3}$ (b) $2.4 \times 10^{10} \text{ m}^{-3}$ (c) $3.6 \times 10^9 \text{ m}^{-3}$ (d) $3.2 \times 10^{10} \text{ m}^{-3}$ [PYQ 2023] Very High

Q4. At equilibrium, in a $p - n$ junction diode the net current is: (a) due to diffusion of majority charge carriers. (b) due to drift of minority charge carriers. (c) zero as diffusion and drift currents are exactly equal and opposite. (d) zero as no charge carriers cross the junction. [PYQ 2020] Very High

Q5. In an n -type semiconductor, the donor energy level theoretically lies: (a) at the exact centre of the energy gap (b) just below the conduction band (c) just above the valence band (d) in the conduction band [PYQ 2020] High

Q6. In a p -type semiconductor, the acceptor energy level theoretically lies: (a) at the centre of the energy gap (b) just below the conduction band (c) just above the valence band (d) inside the valence band [Practice] High

Q7. Assertion (A): The electrical conductivity of an intrinsic semiconductor strictly increases on doping. **Reason (R):** Doping always increases the number of free electrons in the semiconductor. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [CBSE SQP 2022] Very High

Q8. Assertion (A): The resistance of a semiconductor heavily decreases with an increase in temperature. **Reason (R):** At higher temperatures, more covalent bonds break, generating more electron-hole pairs for conduction. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] Very High

Q9. How does an increase in doping concentration specifically affect the width of the depletion layer of a $p - n$ junction diode? [PYQ 2021] Very High

Q10. In a half-wave rectification circuit, what is the exact output frequency if the input AC frequency is **25 Hz**? [PYQ 2021] High

- Q11.** In a full-wave rectifier circuit operating from a **50 Hz** mains supply, what will be the fundamental frequency of the output ripple? [Practice] Very High
- Q12.** What are the typical values of the threshold (cut-in) voltage for Silicon and Germanium diodes? [Delhi Gov. SQP 2022] High
- Q13.** An ideal **$p - n$** junction diode is connected in series with a **100 Ω** resistor and a **5 V** battery in forward bias. What is the current flowing in the circuit? [Practice] Medium
- Q14.** In a pure semiconductor crystal of Si, if antimony is added, what specific type of extrinsic semiconductor is obtained? [CBSE SQP 2022] High
- Q15.** What is the name of the extrinsic semiconductor formed when pure germanium is actively doped with a trivalent impurity like Indium? [CBSE 2022] High
- Q16.** Define the term 'cut-in voltage' (or knee voltage) in the context of a **$p - n$** junction diode. [Delhi Gov. SQP 2022] Very High
- Q17.** How does the potential barrier of a **$p - n$** junction diode change when it is forward biased? [Practice] Very High
- Q18.** In an unbiased **$p - n$** junction, holes diffuse from the p-region to the n-region primarily because: (a) free electrons in the n-region attract them. (b) they move across the junction by the potential difference. (c) hole concentration in the p-region is substantially higher than in the n-region. (d) all of the above. [Practice] High
- Q19.** What is a 'dopant' in semiconductor physics? Give one example. [PYQ 2017] High
- Q20.** What is the net electrical charge of a **p -type** semiconductor crystal? [Practice] Very High
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2-Mark Questions

- Q21.** Distinguish clearly between 'intrinsic' and 'extrinsic' semiconductors on the basis of charge carriers and doping. [PYQ 2015] Very High
- Q22.** Write two fundamental differences between an **n -type** and a **p -type** semiconductor. [Delhi Gov. SQP 2022] Very High
- Q23.** Draw the distinct energy band diagrams for an **n -type** and a **p -type** semiconductor at a temperature **$T > 0 \text{ K}$** . [PYQ 2023] Very High
- Q24.** Explain briefly why the electrical resistance of a **$p - n$** junction is very low when it is forward biased and extremely high when it is reverse biased. [CBSE 2022] High
- Q25.** Answer the following giving scientific reasons: (A) A **$p - n$** junction diode is easily damaged by a very strong current. (B) Impurities are deliberately added into intrinsic semiconductors. [PYQ 2023] High
- Q26.** Distinguish fundamentally between a metal and an insulator on the explicit basis of their energy band diagrams. [PYQ 2014] Very High
- Q27.** The figure shows a piece of pure semiconductor **S** in series with a variable resistor **R** and a source of constant voltage **V** . Should the value of **R** be increased or decreased to keep the

reading of the ammeter constant, when the semiconductor S is heated? Justify your answer. [CBSE SQP 2022] High

Q28. Draw the V-I characteristics of a $p - n$ junction diode. Briefly explain why the current under reverse bias is almost entirely independent of the applied voltage up to the critical breakdown voltage. [PYQ 2020] Very High

Q29. In the circuit shown below, the battery voltage is 6 V . Two ideal diodes D_1 and D_2 are connected in parallel branches with resistors of $150\ \Omega$ and $75\ \Omega$ respectively. A common resistor of $100\ \Omega$ is in series with the battery. If D_1 is forward biased and D_2 is reverse biased, calculate the current drawn from the battery. [Practice] High

Q30. What is the specific physical cause of the potential barrier in a $p - n$ junction? Does it aid or oppose the flow of majority charge carriers? [PYQ 2017] Very High

Q31. Give two important operational differences between a half-wave rectifier and a full-wave rectifier. [CBSE 2022] High

Q32. Two semiconductor diodes **Ge** and **Si** are connected in parallel to a battery. The cut-in voltages for **Ge** and **Si** are 0.3 V and 0.7 V respectively. If the external circuit applies a forward voltage, which diode conducts first, and what will be the effective voltage drop across the parallel combination? [Practice / Source 404] Very High

3-Mark Questions

Q33. Write the two critical processes (diffusion and drift) that take place during the formation of a $p - n$ junction. Explain how they establish dynamic equilibrium. [PYQ 2017] Very High

Q34. Explain with the help of a suitable diagram the physical formation of the depletion region and the barrier potential in an unbiased $p - n$ junction. [PYQ 2016] Very High

Q35. Explain briefly, with the help of a circuit diagram, how the V-I characteristics of a $p - n$ junction diode are practically obtained in: (A) forward bias, and (B) reverse bias. Draw the distinct shapes of the characteristic curves obtained. [PYQ 2018] Very High

Q36. What is formally meant by 'energy band gap' in a solid? Draw the typical energy band diagrams for a conductor, an insulator, and a semiconductor, clearly showing the band gaps. [CBSE 2022] High

Q37. Explain with the help of a labelled circuit diagram, the working principle of a $p - n$ junction diode as a half-wave rectifier. Draw the corresponding input and output waveforms. [PYQ 2016] Very High

Q38. Define the terms 'depletion layer' and 'barrier potential' for a $p - n$ junction. How does (A) an increase in the doping concentration, and (B) applying a forward bias across the junction, affect the width of the depletion layer? [PYQ 2017] High

Q39. In the given circuit, a Silicon diode ($V_k = 0.7\text{ V}$) and a Germanium diode ($V_k = 0.3\text{ V}$) are connected in series with a 12 V battery and a $5.6\text{ k}\Omega$ resistor. Find the exact value of the

output voltage V_0 across the resistor and the current I flowing in the circuit. [Practice / Source 404] Very High

Q40. Differentiate analytically between **p-type** and **n-type** semiconductors based on: (i) Dopant atoms used, (ii) Majority charge carriers, and (iii) Position of the new impurity energy level in the band diagram. [Practice] High

Q41. A pure semiconductor has an equal electron and hole concentration of $6 \times 10^8 \text{ m}^{-3}$. Doping by a specific impurity increases the electron concentration to $9 \times 10^{12} \text{ m}^{-3}$. (A) Calculate the new hole concentration in the doped semiconductor. (B) Identify the type of the semiconductor formed (**p-type** or **n-type**) and justify. [Practice] Medium

Q42. The graph of potential barrier versus width of the depletion region for an unbiased diode is a specific curve. Draw how this potential barrier graph shifts and changes in width when the diode is (i) Forward Biased, and (ii) Reverse Biased. Justify your drawing. [CBSE SQP 2022] High

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Draw a neatly labelled circuit diagram of a Full-Wave Rectifier using two **p – n** junction diodes. (B) Explain its detailed working principle. (C) Draw the time-varying waveforms for the input AC voltage and the corresponding output DC voltage. (D) What is the specific role of the center-tap in the secondary coil of the transformer used in this circuit? [Practice] Very High

Q44. (A) Write the important processes that occur during the formation of a **p – n** junction diode. (B) With the help of a necessary diagram, explain the term 'barrier potential'. (C) Why does the reverse current in a **p – n** junction heavily depend on the temperature of the junction but remains largely independent of the applied reverse voltage up to the breakdown limit? [PYQ 2015 / 2020] Very High

Q45. Case Study: V-I Characteristics of a Diode The circuit arrangement for studying the V-I characteristics of a diode allows the applied voltage to be changed using a potentiometer. In forward bias, the current increases very slowly until the voltage crosses a certain value called the cut-in voltage (**0.2 V** for Ge, **0.7 V** for Si). After this, the current increases exponentially. In reverse bias, a small, almost constant current (microamperes) flows until the breakdown voltage is reached. (A) The specific voltage at which forward bias current increases rapidly is called: (B) As the forward voltage across the diode increases beyond the cut-in voltage, what physically happens to the effective resistance of the diode? (C) What are the specific charge carriers responsible for the small current measured in the reverse bias condition? (D) Why is a milliammeter used in the forward bias circuit while a microammeter is strictly used in the reverse bias circuit? [Delhi Gov. SQP 2022] Very High

Q46. Case Study: Rectification of AC A rectifier is a device that explicitly converts an alternating current (which periodically reverses direction) into a direct current (which flows in only one direction). This process relies on the unidirectional conduction property of **p – n** junction diodes. A single diode acts as a half-wave rectifier, while two diodes with a center-tapped transformer act as a full-wave rectifier. (A) State the fundamental property of a **p – n** junction diode that makes it suitable for rectification. (B) If the input AC frequency is **50 Hz**,

what will be the output ripple frequency for (i) a half-wave rectifier, and (ii) a full-wave rectifier? (C) Does the output of a full-wave rectifier represent a pure, steady DC voltage? If not, what specific electronic component is used across the output to smooth it out? (D) Why is a full-wave rectifier considered more efficient than a half-wave rectifier? [Practice] High

Q47. (A) Distinguish clearly between intrinsic and extrinsic semiconductors. (B) How is a **p-type** semiconductor formed from a pure silicon crystal? Explain the creation of holes using a schematic lattice diagram. (C) Based on energy band theory, explain why an intrinsic semiconductor behaves as a perfect insulator at absolute zero ($T = 0 \text{ K}$) but gains some conductivity at room temperature. [PYQ 2015] High

Q48. (A) Draw a comprehensive circuit diagram to study the V-I characteristics of a **p – n** junction diode in both forward bias and reverse bias. (B) Plot the typical V-I characteristic curves obtained. (C) From the forward characteristics curve, how would you determine the dynamic resistance of the diode at a given operating current (e.g., at $I = 15 \text{ mA}$)? Write the formula. [PYQ 2018 / Delhi Gov SQP 2022] Very High

Q49. Case Study: Energy Bands in Solids According to Bohr's atomic model, electrons in isolated atoms have discrete energy levels. However, in a solid crystal, atoms are packed closely together, causing the outer electron energy levels to split into continuous bands of energy. The highest filled band is the valence band, and the lowest empty band is the conduction band. The energy gap between them determines the electrical properties of the material. (A) On the explicit basis of the energy band gap (E_g), how do you classify a material as a conductor, an insulator, or a semiconductor? (B) What happens to the energy band gap of a semiconductor when its temperature is increased? (C) Carbon (Diamond), Silicon, and Germanium all belong to Group 14 and have 4 valence electrons. Why are Si and Ge considered semiconductors, whereas Carbon (Diamond) acts as an insulator? [Practice] High

Q50. (A) Why are extrinsic semiconductors overwhelmingly preferred over intrinsic semiconductors in manufacturing electronic devices? (B) In a **p – n** junction, if the p-side is connected to a potential of -2 V and the n-side is connected to a potential of -5 V , state whether the diode is forward biased or reverse biased. Justify your answer. (C) A 10 V ideal battery is connected in series with a $1 \text{ k}\Omega$ resistor and an ideal **p – n** junction diode. Calculate the current flowing in the circuit if the diode is (i) forward biased, and (ii) reverse biased. (D) Explain the concept of 'hole' current in the valence band of a semiconductor. [Practice] Very High

Answer Key

Q1: (b) diffusion of both electrons and holes. (Due to concentration gradients, electrons diffuse $n \rightarrow p$ and holes $p \rightarrow n$, leaving behind immobile ions). **Q2:** (b) **C**. (Carbon has the smallest atomic radius among group 14, hence the highest ionization energy / band gap, $\sim 5.4 \text{ eV}$).

Q3: (c) $3.6 \times 10^9 \text{ m}^{-3}$. (By mass action law,
 $n_e \cdot n_h = n_i^2 \Rightarrow n_e = (1.2 \times 10^{15})^2 / (4 \times 10^{20}) = 1.44 \times 10^{30} / 4 \times 10^{20} = 3.6 \times 10^9 \text{ m}^{-3}$
). **Q4:** (c) zero as diffusion and drift currents are exactly equal and opposite. **Q5:** (b) just below

the conduction band. (Donor electrons require very little energy to jump into the conduction band). **Q6:** (c) just above the valence band. (Electrons from the VB easily jump into acceptor levels, creating holes). **Q7:** (c) A is true but R is false. (Doping with a trivalent impurity increases holes, not electrons). **Q8:** (a) Both A and R are true and R is the correct explanation of A. **Q9:** An increase in doping concentration strictly *decreases* the width of the depletion layer (more dopants provide the required uncovering of space charge over a shorter distance). **Q10:** 25 Hz. (In half-wave rectification, output ripple frequency = input frequency). **Q11:** 100 Hz. (In full-wave rectification, output ripple frequency = $2 \times$ input frequency). **Q12:** Germanium $\approx 0.2\text{V}$ to 0.3V ; Silicon $\approx 0.7\text{V}$. **Q13:** $I = V/R = 5/100 = 0.05\text{ A}$ (or 50 mA). (An ideal forward-biased diode acts as a perfect short circuit with zero voltage drop). **Q14:** *n*-type semiconductor. (Antimony is a pentavalent impurity, providing extra electrons). **Q15:** *p*-type semiconductor. (Trivalent impurities create holes). **Q16:** The minimum forward voltage applied across a diode beyond which the forward current starts increasing rapidly (exponentially). **Q17:** The potential barrier strongly *decreases*. (The applied voltage opposes the built-in barrier potential). **Q18:** (c) hole concentration in the p-region is substantially higher than in the n-region. (Diffusion is driven by concentration gradient). **Q19:** A dopant is a desirable impurity atom intentionally added to a pure semiconductor to modify its electrical conductivity. Example: Phosphorus (pentavalent) or Boron (trivalent). **Q20:** Strictly zero (Neutral). (The crystal as a whole contains an equal number of protons and electrons, even after doping).

Q21: Intrinsic: Pure semiconductor, $n_e = n_h$, low conductivity. Extrinsic: Doped semiconductor, $n_e \neq n_h$, high conductivity tailored by doping. **Q22:** *n*-type: Doped with pentavalent impurity; Electrons are majority carriers. *p*-type: Doped with trivalent impurity; Holes are majority carriers. **Q23:** Draw *n*-type showing Donor level (E_D) just below Conduction Band (E_C). Draw *p*-type showing Acceptor level (E_A) just above Valence Band (E_V). Show majority carriers in the respective bands. **Q24:** In forward bias, the applied voltage opposes the barrier, narrowing the depletion layer and allowing majority carriers to cross easily (low resistance). In reverse bias, it aids the barrier, widening the depletion layer and blocking majority carriers (high resistance). **Q25:** (A) Strong current causes excessive Joule heating ($I^2 R$), which breaks millions of covalent bonds permanently, melting/destroying the crystal structure. (B) Intrinsic semiconductors have conductivity too low for practical use at room temperature. Impurities heavily increase charge carriers. **Q26:** Metal: Conduction and Valence bands overlap (zero band gap), allowing easy electron flow. Insulator: Large energy band gap ($E_g > 3\text{eV}$), so electrons cannot jump from VB to CB. **Q27:** R should be strictly *increased*. Heating S generates more electron-hole pairs, heavily reducing its resistance. To keep the total resistance (and thus current) constant, the external series resistance R must be increased. **Q28:** Draw curve in Q3 quadrant. Reverse current is due exclusively to minority carriers. Since minority carriers are generated thermally and are very few in number, the small reverse voltage sweeps all of them across immediately. Increasing voltage doesn't increase their number, so current saturates. **Q29:** D_1 acts as a short wire; D_2 acts as an open switch. Current flows only through the $150\ \Omega$ resistor. Total resistance = $150 + 100 = 250\ \Omega$. Current $I = V/R = 6/250 = 0.024\text{ A}$ (or 24 mA). **Q30:** Cause: The diffusion of electrons from *n* to *p* and holes from *p* to *n* leaves un-neutralized positive donor ions on the n-side and negative acceptor ions on the p-side, creating an electric field. It *opposes* the flow of majority charge carriers. **Q31:** (1) HWR converts only one

half of the AC cycle; FWR converts both halves. (2) FWR has higher efficiency and double the ripple frequency of HWR. **Q32:** The Ge diode conducts first (at 0.3V). Once it conducts, it holds the voltage drop across the parallel combination to $\approx 0.3\text{V}$. The Si diode (0.7V required) will not turn on. Effective voltage drop is 0.3V .

Q33: Diffusion: Due to concentration gradient, majority electrons move $n \rightarrow p$ and majority holes $p \rightarrow n$. Drift: The resulting immobile space-charge creates an electric field from n to p , which sweeps minority carriers across (electron $p \rightarrow n$, holes $n \rightarrow p$). Equilibrium is reached when Diffusion Current = Drift Current. **Q34:** Diagram showing p -side with negative acceptor ions near the junction, n -side with positive donor ions near the junction. Mark the width as 'Depletion Region'. Show the fictitious battery symbol representing 'Barrier Potential' (V_B) with positive terminal towards n -side. **Q35:** (A) Forward bias: p -side to positive, n -side to negative terminal. Draw circuit with rheostat, voltmeter in parallel, mA in series. Curve in first quadrant showing exponential rise after knee voltage. (B) Reverse bias: p -side to negative, n -side to positive. Circuit uses μA . Curve in 3rd quadrant showing constant small negative current until breakdown. **Q36:** The energy gap (E_g) is the energy difference between the bottom of the conduction band and the top of the valence band. Diagram: Conductor (Overlapping bands), Semiconductor ($E_g < 3\text{eV}$), Insulator ($E_g > 3\text{eV}$). **Q37:** Diagram: AC source $\rightarrow$ Step-down Transformer $\rightarrow$ single diode $\rightarrow$ Load Resistor R_L . Working: During the positive half cycle, diode is forward biased and conducts. During the negative half cycle, it is reverse biased and blocks current. Output is pulsating DC. Draw sine wave for input and separated positive humps for output. **Q38:** Depletion layer: Space-charge region near the junction devoid of mobile carriers. Barrier potential: The potential difference across the depletion layer opposing majority carrier flow. (A) Increase in doping decreases the width. (B) Forward bias decreases the width. **Q39:** The diodes are in series. Total voltage drop across both diodes = $0.7\text{V} + 0.3\text{V} = 1.0\text{V}$. Voltage across resistor $V_0 = 12\text{V} - 1.0\text{V} = 11.0\text{V}$. Current $I = V_0/R = 11.0/5600 \approx 1.96 \times 10^{-3} \text{ A} = 1.96 \text{ mA}$. **Q40: p -type:** (i) Trivalent (Boron, Al), (ii) Holes, (iii) Acceptor level just above Valence Band. **n -type:** (i) Pentavalent (P, As), (ii) Electrons, (iii) Donor level just below Conduction Band. **Q41:** (A) Using mass-action law: $n_e n_h = n_i^2$. Given $n_i = 6 \times 10^8$, $n_e = 9 \times 10^{12}$. $n_h = (6 \times 10^8)^2 / (9 \times 10^{12}) = 36 \times 10^{16} / 9 \times 10^{12} = 4 \times 10^4 \text{ m}^{-3}$. (B) Since $n_e \gg n_h$, the semiconductor formed is explicitly **n -type**. **Q42:** Unbiased: Standard potential hill. (i) Forward Biased: The barrier height strongly decreases and depletion width narrows (draw a lower, narrower hill). (ii) Reverse Biased: The barrier height heavily increases and depletion width widens (draw a taller, wider hill).

Q43: (A) Draw transformer, two diodes (D_1 top, D_2 bottom), center tap grounded, load resistor R_L connected to center tap. (B) Positive half cycle: D_1 FB (conducts), D_2 RB. Negative half cycle: D_1 RB, D_2 FB (conducts). Current through R_L flows in the same direction both times. (C) Input: Continuous sine wave. Output: Continuous positive humps. (D) The center tap provides a zero-potential reference to divide the secondary voltage equally, allowing the two diodes to alternate conduction. **Q44:** (A) Diffusion of majority carriers creates a space-charge region. Drift of minority carriers opposes it until equilibrium. (B) Barrier potential is the built-in electric field potential preventing further diffusion of majority carriers. (C) Reverse current is due to minority carriers generated by thermal agitation. It depends strictly on temperature (bond breaking)

and not on applied voltage because the small voltage is already enough to sweep all available minority carriers. **Q45:** (A) Cut-in voltage (or knee voltage). (B) The effective (dynamic) resistance decreases exponentially. (C) Minority charge carriers (electrons in p-side, holes in n-side). (D) Forward current is carried by abundant majority carriers (large current $\rightarrow$ mA). Reverse current is carried by sparse minority carriers (tiny current $\rightarrow \mu\text{A}$). **Q46:** (A) Unidirectional conduction (conducts highly in forward bias, blocks in reverse bias). (B) (i) HWR = **50 Hz**. (ii) FWR = **100 Hz**. (C) No, the output is pulsating DC. A filter circuit (using a capacitor in parallel or inductor in series) is used to smooth it into steady DC. (D) FWR converts both halves of the AC cycle, avoiding power wastage during the negative half-cycle. **Q47:** (A) Intrinsic: Pure, thermally generated carriers ($n_e = n_h$). Extrinsic: Doped with impurity to increase carriers. (B) Pure Si is doped with trivalent impurity (like Boron). Boron has 3 valence electrons, forming 3 bonds with Si. The 4th bond is incomplete, creating a vacancy called a hole. (C) At **0K**, all electrons are tightly bound in covalent bonds (Valence band is full, Conduction band is completely empty). No thermal energy exists to break bonds. **Q48:** (A) Diagram for FB: battery + to p, - to n. V in parallel to diode, mA in series. Diagram for RB: battery - to p, + to n. V in parallel, μA in series. (B) 1st quadrant: exponential rise. 3rd quadrant: constant low current, then sharp breakdown. (C) Dynamic resistance $r_d = \Delta V / \Delta I$ evaluated around **$I = 15 \text{ mA}$** from the tangent to the curve. **Q49:** (A) Conductor: Overlapping bands ($E_g = 0$). Semiconductor: Small gap ($E_g < 3\text{eV}$). Insulator: Large gap ($E_g > 3\text{eV}$). (B) The band gap decreases slightly. More importantly, thermal energy pushes many electrons across the gap into the conduction band. (C) The bond strength (and thus energy gap) depends on interatomic spacing. Carbon has a very small atomic radius, so its valence electrons are extremely tightly bound, resulting in a large band gap ($\sim 5.4\text{eV}$), making it an insulator. Si and Ge have larger radii and weaker bonds ($\sim 1.1\text{eV}$ and $\sim 0.7\text{eV}$). **Q50:** (A) Extrinsic semiconductors have vastly higher conductivity that can be precisely controlled by adjusting the doping concentration, making them practical for devices. (B) Forward Biased. (-2V is higher/more positive than -5V . Since p-side is at a higher potential than n-side, it is FB). (C) (i) Forward biased: ideal diode acts as a wire. **$I = V/R = 10/1000 = 10 \text{ mA}$** . (ii) Reverse biased: ideal diode acts as an open circuit. **$I = 0 \text{ A}$** . (D) When an electron in the valence band breaks free, it leaves a vacancy. A neighboring valence electron can jump into this vacancy, moving the vacancy. The apparent movement of this positive vacancy is treated as 'hole' current in the valence band.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).